

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 2.2 AREA OF POLYGONS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Answer the driving question: How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available? Remember the shamba left to three siblings in Nakuru County. It has five boundary corners marked by old fence posts, and no two adjacent sides are equal. Use what you learned in all eight lessons. Write your answers in the space under each prompt. Show all working for calculations, include units, and explain your reasoning in full sentences. Use your Growing Model and the Driving Question Board to help you.

Part 1: Why the usual formula is not enough

Surveyor Wanjiku stands at the edge of the five-sided shamba. She cannot simply use $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$ on it. (a) Explain two reasons why this formula cannot be applied directly to the whole shamba. (b) What must be true about the 'height' in this formula? (c) What exactly did the class need to find out to solve the problem?

(a) First, the shamba is an irregular five-sided shape, not a single triangle, so one triangle formula cannot cover it. Second, the formula needs a perpendicular height, and on uneven ground with maize crops (or a stream in the way) the surveyor cannot measure that height. (b) The height must be the perpendicular distance from the base to the opposite vertex. It is not the same as a slanted side length. (c) We needed a way to find the area of a triangle without a perpendicular height, and a way to break the five-sided plot into triangles. This led us to the sine formula, Heron's formula and decomposition into triangles.

Part 2: Choosing and using triangle formulas

Wanjiku's field notes contain two triangular sections. Triangle ABC: AB = 120 m, AC = 95 m, and the angle at A between them is 68° . Northern sub-plot: sides 40 m, 55 m and 63 m, with no angle or height recorded. (a) For each triangle, name the formula you would use and explain why it is the right choice. (b) Calculate the

(a) Triangle ABC: two sides and the included angle are known, so use $A = \frac{1}{2}ab \sin C$. Northern sub-plot: only the three side lengths are known, with no angle or height, so use Heron's formula. (b) Triangle ABC: $A = \frac{1}{2} \times 120 \times 95 \times \sin 68^\circ = 5\,700 \times 0.9272 \approx 5\,285 \text{ m}^2$. Northern sub-plot: $s = (40 + 55 + 63) \div 2 = 79 \text{ m}$. $A = \sqrt{[79 \times (79 - 40) \times (79 - 55) \times (79 - 63)]} = \sqrt{[79 \times 39 \times 24 \times 16]} = \sqrt{1\,183\,104} \approx 1\,088 \text{ m}^2$. (c) The sine formula comes from $h = b \sin \theta$, where the height is found from one side and the angle that side makes with the base. That only works if the angle lies between the two known sides, because then one of those sides is the base and the other is the slanted side from which the height is found.

area of each triangle. Give answers to the nearest square metre. (c) Why must the angle in the sine formula be the included angle?	
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Part 3: Breaking the five-sided plot into triangles

(a) How many triangles does the five-sided shamba break into when you draw diagonals from one vertex? Which diagonals would you draw from vertex A, and what are the triangles called? (b) State the rule for any polygon with n sides. (c) Explain how you would then find the total area, and say how you would choose the formula for each triangle.	(a) It breaks into 3 triangles. Draw the diagonals AC and AD from vertex A. The triangles are ABC, ACD and ADE. (b) A polygon with n sides can be split into $(n - 2)$ non-overlapping triangles by drawing diagonals from one vertex. Here $5 - 2 = 3$. (c) Find the area of each triangle and add them to get the total area of the shamba. For each triangle, look at the measurements in the field notes: use $\frac{1}{2}ab \sin C$ if two sides and the included angle are known, Heron's formula if all three sides are known, and $\frac{1}{2} \times \text{base} \times \text{height}$ if a perpendicular height is available.
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Part 4: Testing ideas and spotting mistakes

(a) A learner says: 'Longer sides always give a larger area, so a triangle with sides 60 m and 90 m must have a bigger area than any triangle with shorter sides, whatever the angle between the sides.' Rewrite this into a clear correct statement about two triangles that have the same two sides but different included angles. Then calculate the area of a triangle with sides 60 m and 90 m and an included angle of 30°, and explain what this shows about the effect of the included angle. (b) Why can the regular polygon formula $A = ns^2 \div [4 \tan(180^\circ/n)]$ not be used on Wanjiku's shamba?	(a) A correct statement: longer sides alone do not guarantee a larger area. For two triangles with the same two side lengths, the one with the larger included angle (up to 90°) has the larger area, because the area depends on $\sin \theta$. For 60 m, 90 m and 30°: $A = \frac{1}{2} \times 60 \times 90 \times \sin 30^\circ = 2\,700 \times 0.5 = 1\,350 \text{ m}^2$. This shows that if the included angle is small, the area is small even when the sides are long. (b) The regular polygon formula only works when all sides are equal and all angles are equal, so the polygon splits into identical central triangles. The shamba has unequal sides and unequal angles, so it must be decomposed into triangles instead.
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Part 5: Your final explanation

Answer the driving question in a complete explanation: How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available? Your answer should (a) describe the steps the surveyor follows from start to finish, (b) name the formulas in her toolkit and say when each one is used, (c) explain how the area would be converted for practical land use, using Triangle ABC as an example, and (d) say how your thinking has changed since Lesson 1.	(a) The surveyor first sketches the plot and draws diagonals from one vertex, splitting an n-sided plot into $(n - 2)$ triangles. For the five-sided shamba these are ABC, ACD and ADE. For each triangle she checks which measurements she has, chooses a formula, calculates its area, and adds all the areas to get the total. (b) Toolkit: $\frac{1}{2} \times \text{base} \times \text{height}$ when a perpendicular height is available; $A = \frac{1}{2}ab \sin C$ when two sides and the included angle are known; Heron's formula, $A = \sqrt{s(s - a)(s - b)(s - c)}$ with $s = (a + b + c) \div 2$, when all three sides are known and there is no angle or height; specific quadrilateral formulas for parallelograms, trapezoids and kites; and $A = ns^2 \div [4 \tan(180^\circ/n)]$ for regular polygons only. Because the sine formula and Heron's formula do not need a perpendicular height, she can solve the plot even when the stream or crops stop her from measuring heights. (c) Areas are in square metres. Since 1 hectare = 10 000 m², Triangle ABC of about 5 285 m² is about 0.53 hectares, and the whole shamba is found by converting the total in the same way. (d) In Lesson 1 I thought we needed a height for every triangle and I could not see how to start. Now I know that decomposing the plot and choosing the formula to match the data lets the surveyor find the area of any polygon.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the limitation of $\frac{1}{2} \times \text{base} \times \text{height}$	Clearly explains that a perpendicular height is needed, that it cannot be measured on the shamba, and links this to the need for new methods.	States that the height is needed and cannot be measured but gives little link to the new methods.	Gives a vague or incorrect reason, such as confusing the height with a side length.
Selecting and applying formulas correctly	Chooses the correct formula for each triangle with a justification based on the data, calculates correctly with units and sensible rounding.	Chooses the correct formulas and calculates mostly correctly, with a small slip in working, units or justification.	Chooses the wrong formula or makes major calculation errors.
Decomposing polygons	States the $(n - 2)$ rule, names the diagonals and triangles for the shamba, and explains summing the areas.	Gives the number of triangles and the summing idea but omits some detail, such as the diagonals or the rule.	Does not decompose the polygon, or gives an incorrect number of triangles.
Scientific explanation of the driving question	Gives a complete, logical explanation connecting all steps, the toolkit, unit conversion and personal growth in understanding.	Explains most steps and tools but leaves out unit conversion or reflection.	Gives a list of formulas with no connection to the surveyor's problem.

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